

HomeCell S8 Home Battery: Owner and Installer Manual

Solvane Energy Ltd. Manual version 4.1. This manual covers the HomeCell S8 home battery system, model number HC-S8-10, and the optional HomeCell X2 expansion module.

About Solvane Energy

Solvane Energy was founded in 2016 in Halden, Norway, by engineers Ingrid Vale and Tomas Brekke. The company designs battery storage systems for homes and small businesses. Today Solvane has about 640 employees and sells its products in 14 countries.

About this manual

Read this manual fully before you install or use the battery. Chapters 1 to 9 are for installers. Chapters 10 to 18 are for everyday owners. Chapters 19 to 28 cover error codes, troubleshooting, warranty and support. Keep this manual near the battery for future reference.

Important words: Warning means there is a risk of injury. Caution means there is a risk of damage to the product. Note means extra helpful information.

Chapter 2: Safety Warnings

The HomeCell S8 stores a large amount of electrical energy. Wrong installation or use can cause electric shock, fire or burns.

Warnings

Only a licensed electrician may install, open or wire the battery. Never open the front cover while the system is switched on. Wait at least 5 minutes after switching off before touching any terminal, because capacitors inside stay charged.

Do not install the battery in a bedroom, a bathroom or any room where people sleep. Keep children and pets away from the unit.

If you smell burning or see smoke, do not try to put out the fire with water. Leave the building and call the fire service. Tell them the building has a lithium iron phosphate (LFP) battery.

Cautions

Do not stack objects on top of the battery. Do not cover the air vents on the sides. Do not spray water or cleaning liquid on the unit. Do not drop the battery during transport; a dropped unit must be checked by Solvane before use, even if it looks fine.

Chapter 3: What Is in the Box

Check that every part below is in the box before you start. If something is missing or damaged, contact your installer.

Part number	Item	Quantity
HC-S8-10	HomeCell S8 battery unit	1
HC-WB2	Steel wall bracket	1
HC-FK8	Floor stand kit (optional mounting)	1
HC-CT40	Current sensor clamp, 40 mm	2
HC-AC6	AC connector plug	1
HC-DC4	Solar (PV) connector pair	2
HC-BLT8	M8 anchor bolts	6
HC-QG	Quick start guide	1

Note: The current sensor clamps (HC-CT40) measure how much power your home uses. Without them, the battery cannot follow your home's demand and will only run in Backup mode.

Keep the box and foam for at least 30 days. You will need them if the unit has to be returned.

Chapter 4: Technical Specifications

Item	Value
Model	HC-S8-10
Usable capacity	10.2 kWh
Battery chemistry	Lithium iron phosphate (LFP)
Continuous output power	5.0 kW
Peak output power (10 seconds)	7.5 kW
Round-trip efficiency	94.5%
Weight	98 kg
Size (H x W x D)	1050 x 600 x 210 mm
Noise level	Below 30 dB
Protection rating	IP65
Operating temperature	-10 C to 50 C
Design life	6000 cycles

Note: Usable capacity is measured at 25 C. In cold weather the battery may deliver less energy. Up to three HomeCell X2 expansion modules can be added; see Chapter 23.

Chapter 5: Choosing a Location

The right location keeps the battery safe and helps it last longer.

Indoors or outdoors

The HomeCell S8 has an IP65 rating, so it can be installed outdoors. Outdoors, choose a wall that is shaded in the afternoon. Direct sun can push the battery above its working temperature and reduce power. A garage or utility room is the best indoor location.

Distances

Leave at least 30 cm of free space on the left and right sides for air flow, and 50 cm above the unit. Keep the battery at least 1 metre away from gas meters, boilers, heaters and any fuel or paint storage.

Install the battery so that its bottom edge is at least 15 cm above the floor, to protect it from flooding and splashing water.

Wall strength

The wall must carry at least four times the weight of the battery. Brick and concrete walls are suitable. Plasterboard or wooden walls are not suitable; use the floor stand kit (HC-FK8) instead.

Chapter 6: Wall Mounting

Two people are needed to lift the battery. Use the lifting handles on the sides only.

Steps

1. Hold the steel wall bracket (HC-WB2) against the wall and check it is level with a spirit level. 2. Mark the six holes and drill them 12 mm wide and 90 mm deep. 3. Insert the M8 anchor bolts (HC-BLT8) and tighten them to a torque of 25 Nm. 4. Lift the battery and hook it onto the top edge of the bracket. 5. Fit the two safety screws at the bottom of the bracket so the battery cannot be lifted off.

Caution: Do not tighten the anchor bolts above 25 Nm. Too much torque can crack brick and weaken the hold.

Floor stand

If the wall is not strong enough, use the floor stand kit (HC-FK8). The stand must still be fixed to the wall with the two top straps so that the battery cannot tip over.

Chapter 7: Electrical Wiring (AC Side)

Warning: Only a licensed electrician may do this work. Switch off the main breaker before you start.

AC cable

Use copper cable with a cross-section of at least 6 mm squared for the AC connection. The cable must not be longer than 20 metres. Connect the cable to the AC connector plug (HC-AC6): L to the brown terminal, N to the blue terminal and PE to the yellow-green terminal.

Breaker

Install a dedicated 32 A Type B circuit breaker for the battery in your main panel. Do not share this breaker with other devices.

Current sensor clamps

Place the current sensor clamps (HC-CT40) around the incoming grid cables, after the electricity meter. The arrow on the clamp must point toward the house. If the arrow points the wrong way, the battery will charge when it should discharge.

Backup circuit

Loads you want powered during an outage, such as the fridge and lights, must be wired to the BACKUP terminal. The backup circuit can supply up to 5.0 kW.

Chapter 8: Connecting Solar Panels

The HomeCell S8 has a built-in solar charger with two inputs, called PV1 and PV2. You can connect solar panels directly.

Limits

Maximum input voltage per input: 550 V DC. Maximum input current per input: 14 A. Maximum total solar power: 8 kW. Exceeding the maximum input voltage will permanently damage the unit and is not covered by the warranty.

Steps

1. Cover the solar panels or switch off the DC isolator. 2. Measure the voltage of each string with a multimeter and check the polarity. 3. Fit the solar connector pairs (HC-DC4) to the cables. 4. Connect the plus cable to PV+ and the minus cable to PV-. 5. Switch on the DC isolator only after the AC side is finished.

Note: If you already have a separate solar inverter, do not connect the panels to PV1 and PV2. Instead, the battery will charge from the AC side. This is called AC coupling.

Chapter 9: First Start-Up

Before you switch on the battery for the first time, check the following list.

Checklist

1. All covers are closed and all screws are tight. 2. The AC cable is connected and the 32 A breaker is still off. 3. The current sensor clamps are fitted with the arrow pointing toward the house. 4. The solar strings are tested and the DC isolator is still off. 5. Nothing is blocking the air vents.

Start-up order

First switch on the 32 A AC breaker. Then press the black power button on the right side of the battery for 3 seconds. The LED will glow white while the system runs a self-test. The self-test takes about 2 minutes. When the LED turns steady green, switch on the DC isolator for the solar panels.

If the LED turns red during the self-test, read the error code in the app or on the display, and see Chapters 19 to 21.

Chapter 10: Setting Up the HomeCell App

The HomeCell app lets you see your battery, solar and home use, and change settings. It is free for iPhone and Android.

Create an account

Download the HomeCell app from your app store. Tap Sign Up and enter your email address. You will receive a 6-digit code by email; enter it to confirm your account.

Add your battery

Tap the plus sign and choose Add Device. Scan the QR code on the label on the right side of the battery. The label also shows the serial number, which starts with SV8. If the QR code cannot be read, type the serial number by hand.

Share access

You can share your battery with family members. Go to Settings, then Users, then Invite. Shared users can see data, but only the owner can change modes and schedules.

Chapter 11: Wi-Fi and Network Connection

The battery needs an internet connection for the app, firmware updates and remote support.

Connecting to Wi-Fi

In the app, open Settings, then Network, then Wi-Fi. Pick your home network and type the password. The battery only supports 2.4 GHz Wi-Fi networks. It cannot connect to 5 GHz networks. If your router uses one name for both, ask your internet provider how to create a separate 2.4 GHz network.

Using a cable

For a more stable connection, connect an Ethernet cable to the LAN port under the bottom cover. A cable connection is recommended if the battery is installed in a garage far from the router.

Firewall

The battery talks to the Solvane cloud on port 8883. If your company or school network blocks this port, the app will show the battery as offline.

Chapter 12: Operating Modes

You can choose one of four modes in the app under Settings, then Mode.

Self-use mode (default)

The battery stores extra solar energy during the day and uses it in the evening and at night. This mode gives the highest use of your own solar power and the lowest electricity bill for most homes.

Backup mode

The battery stays charged to 100% at all times and only discharges during a power outage. Choose this mode if outages are common where you live.

Time-of-use mode

The battery charges when electricity is cheap and discharges when it is expensive. You set the times yourself; see Chapter 14.

Eco mode

Eco mode works like Self-use, but it keeps at least 20% charge in reserve for outages and limits the charge to 90% to make the battery last longer. Eco mode is recommended if you want a longer battery life.

Chapter 13: Backup Power During Outages

When the grid power goes off, the HomeCell S8 switches the backup circuit to battery power in less than 20 milliseconds. Most devices, such as computers and TVs, will not turn off.

What gets power

Only devices connected to the BACKUP terminal get power during an outage. Other circuits in your home will be off. Your installer decides which circuits are on the backup terminal.

How long it lasts

A full battery (10.2 kWh) can run a fridge, lights, Wi-Fi router and a TV for about 18 to 24 hours. Heaters, ovens and kettles use much more power and shorten this time a lot.

Solar during an outage

If you have panels on PV1 and PV2, they keep charging the battery during an outage in daylight. Panels connected to a separate solar inverter (AC coupling) may stop working during an outage, by law in many countries.

When the grid comes back, the battery switches back automatically. You do not need to do anything.

Chapter 14: Time-of-Use Schedules

Many electricity companies charge less at night. A time-of-use schedule lets the battery fill up with cheap power and use it later.

Create a schedule

Set the mode to Time-of-use (see Chapter 12). Then go to Settings, then Schedules, then New. For each period, choose a start time, an end time and an action: Charge from grid, Discharge, or Hold. You can create up to 6 periods per day.

Example

Charge from grid from 01:00 to 05:00 when power is cheap. Hold from 05:00 to 16:00. Discharge from 16:00 to 21:00 when power is most expensive.

Grid charge limit

You can limit how full the battery charges from the grid, for example to 80%, so there is still room for solar energy the next morning. The setting is called Grid Charge Limit.

Note: Schedules use the time zone set in the app. Check it after daylight saving time changes.

Chapter 15: Display and LED Lights

The LED strip on the front shows the battery status at a glance.

LED	Meaning
Steady white	Starting up / self-test running
Steady green	Working normally
Breathing green	Charging
Steady blue	Discharging to the home
Blinking amber	Firmware update in progress, do not switch off
Steady amber	Warning, battery still works, check the app
Steady red	Fault, battery has stopped, check the error code
Off	Battery is off or in deep sleep

Small display

Press the round button under the LED strip once to show the charge level (SOC) on the small display. Press it twice to show the current power in kW. Press and hold for 5 seconds to show the last error code.

Chapter 16: Firmware Updates

Solvane releases firmware updates to improve performance and add features. Updates are free.

Automatic updates

By default, updates install automatically between 02:00 and 04:00 at night. During an update the LED blinks amber. Do not switch off the battery during an update; this can damage the control board.

Manual updates

To update now, open the app, go to Settings, then About, then Check for Updates. The battery must be at least 20% charged.

Version history

Firmware 3.0 added Time-of-use mode. Firmware 3.1 fixed a problem where the battery showed the wrong charge level in cold weather. Firmware 3.2 added Eco mode and support for the HomeCell X2 expansion module. Firmware 3.3 shortened the backup switch time to under 20 milliseconds.

To see your current version, go to Settings, then About.

Chapter 17: Cleaning and Maintenance

The HomeCell S8 needs very little maintenance.

Cleaning

Every 3 months, wipe the outside with a dry, soft cloth. Use a slightly damp cloth for dirt, but never spray water or use cleaning products, solvents or a pressure washer.

Air vents

Check the air vents on both sides every 6 months. Remove dust, leaves and spider webs with a soft brush or a vacuum cleaner. Blocked vents make the battery run hot and lose power.

Yearly check

Once a year, ask your installer to check that all cables are tight, the wall bracket is secure, and no water has entered the bottom cover.

Long periods away

If you leave your home for more than a month, keep the battery switched on. A battery that is off for a long time can discharge too much and be damaged.

Chapter 18: Battery Care and Temperature

Temperature has a large effect on battery life and power.

Cold weather

Below 0 C the battery reduces charging power to protect the cells. Below -10 C it stops charging completely. The built-in heater warms the cells automatically when the battery is connected to power, so you do not need to do anything. Do not use a space heater or heat gun near the battery.

Hot weather

Above 45 C the battery reduces output power. Above 50 C it stops to protect itself and shows error E131. Make sure the battery is shaded and the air vents are clear.

Best practice

The battery lasts longest when it is kept between 15 C and 30 C and not left at 100% charge for many weeks. Eco mode helps with this. After 6000 cycles the battery still keeps at least 70% of its capacity.

Chapter 19: Error Codes E100 to E199 (Battery Cells)

If the LED turns amber or red, the app shows an error code. Codes E100 to E199 are about the battery cells.

Code	Meaning	What to do
E101	Cell voltage too low	Charge from grid; call support if it repeats
E108	Cell voltage too high	Restart; call support if it repeats
E115	Cell voltage difference too large	Wait 24 h for automatic balancing
E122	Cell temperature too low	Wait for the heater; check location
E131	Cell temperature too high	Shade the unit; clean the air vents
E142	Battery management board not responding	Restart; call support if it repeats
E156	Insulation fault in battery pack	Switch off; call support now
E173	Expansion module not detected	Check X2 cable; see Chapter 23
E188	Battery capacity below 70%	Contact support about warranty

To restart the battery, switch off the 32 A breaker, wait 5 minutes, then switch it on and press the power button for 3 seconds.

Chapter 20: Error Codes E200 to E299 (Inverter and Grid)

Codes E200 to E299 are about the inverter, the grid connection and the solar inputs.

Code	Meaning	What to do
E201	Grid voltage too high	Usually a grid problem; wait
E204	Grid voltage too low	Usually a grid problem; wait
E209	Grid frequency out of range	Wait; check with your power company
E215	Current sensor clamp reversed	Electrician must turn the clamp around
E222	Current sensor clamp not detected	Check the clamp cable
E236	PV input voltage too high	Switch off DC isolator now; call installer
E241	PV polarity reversed	Switch off DC isolator; installer must fix
E257	Inverter overheating	Clean the air vents; reduce load
E268	Backup circuit overload	Turn off some backup devices
E290	Inverter internal fault	Switch off; call support

Note: Grid errors (E201, E204, E209) usually clear by themselves when the grid is stable again. If they appear every day, ask your power company to check the voltage at your home.

Chapter 21: Error Codes E300 to E399 (Network and App)

Codes E300 to E399 are about communication. The battery keeps working normally with these errors, but the app may not show live data.

Code	Meaning	What to do
E301	Wi-Fi password wrong	Enter the password again in the app
E304	Wi-Fi signal too weak	Move the router closer or use a cable
E307	Cannot reach Solvane cloud	Check internet; allow port 8883
E312	5 GHz network selected	Choose a 2.4 GHz network
E318	Clock not set	Connect to the internet to set the time
E325	Firmware download failed	It retries next night automatically
E333	App account not linked	Add the battery again in the app
E349	Meter communication lost	Check the meter cable

Network errors do not affect backup power. Even without internet, the battery protects your home during an outage.

Chapter 22: Troubleshooting Common Problems

The charge level stays at 99% and never reaches 100%

This is normal. Once a week the battery does a full balance charge to 100%. On other days it stops at 99% to reduce wear.

The battery charges from the grid when I do not want it to

Check the mode. In Time-of-use mode, a Charge from grid period may be set. Also check that the current sensor clamp arrow points toward the house (error E215).

The battery does not discharge in the evening

In Backup mode the battery only discharges during outages. In Eco mode it stops at 20%. Check the mode in the app.

The app shows the battery as offline

Check your Wi-Fi and look for error codes E301 to E318. Restarting your router often helps.

The battery makes a clicking sound

A short click when switching between grid and battery is normal. It is the relay. A constant buzzing is not normal; contact support.

Solar production is lower than expected

Check for shade on the panels, dirty panels, and error codes E236 or E241. In very hot weather the battery may limit charging (see Chapter 18).

Chapter 23: Expansion Modules (HomeCell X2)

The HomeCell X2 (part number HC-X2-5) is an expansion module that adds 5.1 kWh of usable capacity. It does not add more output power; the maximum output stays at 5.0 kW.

How many

You can connect up to three HomeCell X2 modules to one HomeCell S8, for a total of 25.5 kWh. The battery must run firmware 3.2 or newer to support expansion modules.

Installing a module

1. Switch off the battery and the 32 A breaker, and wait 5 minutes. 2. Mount the module directly below the HomeCell S8 using its own bracket. 3. Connect the X2 link cable to the EXP port on the bottom of the S8. 4. Switch on. The S8 finds the module automatically within 10 minutes.

If the module is not found, the app shows error E173. Check that the X2 link cable is pushed in until it clicks.

Note: Modules added more than 2 years after the S8 was installed get their own separate warranty.

Chapter 24: Warranty

Solvane Energy gives a limited warranty on the HomeCell S8 and HomeCell X2.

Warranty period

The warranty lasts 10 years from the installation date, or until the battery has delivered 30 MWh of energy, whichever comes first. During the warranty, the battery will keep at least 70% of its usable capacity.

What is covered

Faults in materials and workmanship, and capacity falling below 70% within the warranty period.

What is not covered

Damage from installation by someone who is not a licensed electrician; damage from floods, fire or lightning; solar input voltage above 550 V; opening the unit without permission from Solvane; and normal cosmetic wear such as scratches.

Registering

Your installer must register the battery in the Solvane installer portal within 60 days of installation. If the battery is not registered, the warranty is reduced to 5 years.

Chapter 25: Returns and Repairs (RMA)

If your battery needs repair or replacement, you must get a Return Material Authorization (RMA) number from Solvane before sending anything.

How to get an RMA number

Ask your installer to open a case in the Solvane installer portal, or contact Solvane support (see Chapter 28). Have your serial number (starting with SV8) and the error code ready. Solvane will reply within 2 working days.

Sending the unit

Write the RMA number clearly on the outside of the box. Use the original box and foam if possible. Solvane pays the shipping cost for units under warranty. Units sent without an RMA number will be returned to the sender unopened.

Repair time

Most repairs are finished within 15 working days after Solvane receives the unit. If the repair takes longer, Solvane may send a replacement unit.

Refunds

If you bought the battery directly from Solvane, you can return an unused, unopened unit within 30 days for a full refund.

Chapter 26: Recycling and Disposal

Do not throw the battery away with household waste. Lithium batteries must be recycled.

Free take-back

At the end of its life, Solvane will collect your HomeCell S8 for free and send it to a certified recycling partner. To book a collection, use the End of Life form in the app under Settings, then Support.

What is recycled

More than 90% of the battery materials can be recovered, including lithium, iron, copper and aluminium. The steel case and the plastic parts are recycled separately.

Before collection

Your installer must disconnect the battery from the AC and solar cables. The battery should be charged to between 20% and 30% before collection, because this is the safest level for transport.

Note: Damaged or swollen batteries must not be moved by you. Contact Solvane support for special collection.

Chapter 27: Glossary

Term	Meaning
SOC	State of charge: how full the battery is, in percent
SOH	State of health: battery capacity now compared to when new
kWh	Kilowatt-hour: a unit of energy
kW	Kilowatt: a unit of power
PV	Photovoltaic: solar panels
LFP	Lithium iron phosphate: the safe battery chemistry used in HomeCell
AC coupling	Charging the battery from the AC side, from a separate solar inverter
Cycle	One full charge and one full discharge
Grid	The public electricity network
RMA	Return Material Authorization: a number needed to return a unit
BMS	Battery management system: the board that controls and protects the cells
IP65	Protection against dust and water jets

Chapter 28: Contact and Support

If this manual does not solve your problem, contact Solvane support.

Phone

Call +47 21 55 80 00, Monday to Friday, 08:00 to 18:00 Central European Time. For urgent safety problems, such as error E156, the line is open 24 hours a day.

Email and chat

Email support@solvane.example or use the chat in the HomeCell app under Settings, then Support. Email replies take up to 1 working day.

Before you contact us

Please have this information ready: the serial number (starts with SV8), the firmware version, the error code, and a short description of what happened.

Installer support

Installers have a separate phone line, +47 21 55 80 10, and can open cases in the Solvane installer portal.

Thank you for choosing Solvane Energy.